

1 Compute Latitude θ and Longitude ϕ

READ THIS TUTORIAL ONLY IF YOU HAVE READ THE TUTORIAL ON TEXTURE MAPPING ON A BLOCK! The former explains the motivation and rationale behind texture mapping in general.

Given a surface point $\mathbf{P}$, the outward unit normal is relative to the center $\mathbf{C}$ is:

$$\mathbf{N} = \frac{\mathbf{P} - \mathbf{C}}{R}$$

The conversion from spherical to Cartesian coordinates is given by:

$$x = r \sin \phi \cos \theta$$

$$y = r \sin \phi \sin \theta$$

$$z = r \cos \phi$$

and vice versa:

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \cos^{-1} \frac{z}{r}, \quad \phi = \tan^{-1}(y, x) \quad (1.1)$$

As you can see in the figure below, $\theta \in [-\pi, \pi]$ is called *azimuthal angle* or *longitude* and $\phi \in [0, \pi]$ is called *polar angle* or *latitude*.

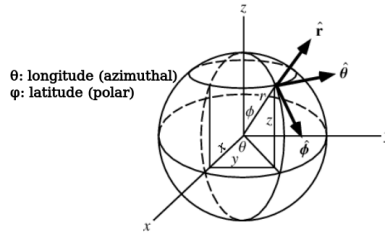


Fig. 1. Spherical coordinate terms

However in ray tracing, we normally know the normal vector $\mathbf{N} = (N_x, N_y, N_z)$, $\|\mathbf{N}\| = 1$ on the sphere so we want to express θ, ϕ given this. The normal is therefore written as $\mathbf{N} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \phi)$. Then we can easily recover the angles from it as:

$$\phi = \cos^{-1}(N_z), \quad \theta = \tan^{-1}\left(\frac{N_y}{N_x}\right)$$

2 Map Them To Texture Coordinates

We know that $\theta \in [-\pi, \pi]$ and $\phi \in [0, \pi]$ therefore to convert them in normalized texture coordinates $u, v \in [0, 1]$, we linearly map into $[0, 1]$ in the same fashion as in block texture mapping.:

$$u = \frac{\theta}{2\pi} + \frac{1}{2}, \quad v = \frac{\phi}{\pi}.$$

If the texture uses the convention that $v = 0$ is the top row, one may flip

$$v \leftarrow 1 - v.$$

Just like with block texture mapping, if we want to repeat the texture r_u times over full θ and r_v times over full ϕ , we multiply by the repeat factor and then wrap (keep only the fractional part);

$$\begin{aligned} u &\leftarrow u \cdot r_u, & v &\leftarrow v \cdot r_v \\ u &\leftarrow u - \lfloor u \rfloor, & v &\leftarrow v - \lfloor v \rfloor \end{aligned}$$

Next, if the texture image has some width and height t_w, t_h , we multiply by $t_w - 1, t_h - 1$ to find the pixel index the (u, v) mapping corresponds to:

$$x = u(W - 1), \quad y = v(H - 1),$$

We can make this more accurate with nearest-neighbor or bilinear sampling.

Notice that there are some edge cases:

- **Poles**; at the north $(0, 1, 0)$ and south $(0, -1, 0)$ poles, longitude is undefined: $\tan^{-1}(0, 0)$ is not meaningful. Any u value is allowed, and implementations typically set $u = 0$ or reuse the u from the previous ray.
- **Seam**; try to ensure the texture sampler wraps horizontally to avoid discontinuities at $u = 0/1$.

Pseudocode

Algorithm 1 Texture Mapping on a Sphere (Equirectangular Mapping)

Require: $\mathbf{P}$ (hit point), sphere of center $\mathbf{C}$ and radius R , texture image $\mathbf{T} \in \mathbb{R}^{H \times W \times 3}$ of H rows and W columns.

Ensure: Texture color at hit point

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1: procedure  $(t)_{est}$ 
2:    $\mathbf{N} \leftarrow \frac{\mathbf{P} - \mathbf{C}}{R}$                                 ▷ NORMAL RELATIVE TO SPHERE'S ORIGIN
3:    $\theta \leftarrow \tan^{-1}(N_z/N_x)$                                 ▷  $[-\pi, +\pi]$  LONGITUDE
4:    $\phi \leftarrow \cos^{-1}(N_y)$                                 ▷  $[0, \pi]$  LATITUDE
5:    $u \leftarrow \frac{\theta}{2\pi} + \frac{1}{2}$ 
6:    $v \leftarrow \frac{\phi}{\pi}$ 
7:   IF FLIP_V THEN                                ▷ IF TRUE,  $v = 0$  CORRESPONDS TO TOP ROW
8:      $v \leftarrow 1 - v$ 
9:    $u \leftarrow u \cdot \text{REPEAT\_U}$ 
10:   $v \leftarrow v \cdot \text{REPEAT\_V}$ 
11:   $u \leftarrow u - \lfloor u \rfloor$                                 ▷ WRAP HORIZONTALLY
12:   $x \leftarrow u \cdot (W - 1)$ 
13:   $y \leftarrow v \cdot (H - 1)$ 
14:  RETURN  $\mathbf{T}[y, x]$ 

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2.1 Numerical Example

Let $\mathbf{P} = (0.6, 0.5, 0.64)$. Notice that $\|\mathbf{P}\| \approx 1$ so we don't need to normalize it ($\mathbf{N} = \mathbf{P}$).

$$\theta = \tan^{-1}(0.64/0.6) \approx 0.82 \text{ rad}, \quad \phi = \cos^{-1}(0.5) = \frac{\pi}{3}.$$

Thus

$$u = \frac{0.82}{2\pi} + \frac{1}{2} \approx 0.63, \quad v = \frac{\pi/3}{\pi} = 0.333.$$

For a 1024×512 texture:

$$x \approx 0.63 \cdot 1023 \approx 644, \quad y \approx 0.333 \cdot 511 \approx 170.$$